Plasticity, nutrition, and breeding

Fecundity as a linear function

First, we model external conditions as a normal distribution

$$C(y) = \frac{1}{\sqrt{2\pi w^2}} \exp\left(-\frac{y^2}{2w^2}\right)$$

And fecundity as a linear function:

$$F(n, y) = a_f n + cy.$$

Then, the fitness is given by:

$$W(n, y) = C(y)F(n, y) = \frac{1}{\sqrt{2\pi w^2}} \exp\left(-\frac{y^2}{2w^2}\right)(a_f n + cy)$$

To find the optimal breeding date, we take the partial derivative against y:

$$\frac{\partial W(n, y)}{\partial y} = \frac{1}{\sqrt{2\pi w^2}} \exp\left(-\frac{y^2}{2w^2}\right) \times (c - y(a_f n + cy)/w^2)$$

Setting this euqal to zero, we have:

$$y = \frac{-a_f n + \sqrt{(a_f n)^2 + 4c^2 w^2}}{2c}$$

Now, if we average the fitness $W(n, y)$ over the breeding date distribution y, we get the mean fitness:

$$
\begin{aligned}
\bar{W}(n,\bar{y}) &= \int p(y)W(n,y)dy \\
&= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma_y^2}} \exp\left(-\frac{(y-\mu)^2}{2\sigma_y^2}\right) \times \frac{1}{\sqrt{2\pi w^2}} \exp\left(-\frac{y^2}{2w^2}\right)(a_f n + cy)dy \\
&= \frac{1}{2\pi\sigma_y\omega} \int_{-\infty}^{\infty} \exp\left(-\frac{(y-\mu)^2}{2\sigma_y^2}\right) \times \exp\left(-\frac{y^2}{2w^2}\right)(a_f n + cy)dy \\
&= \frac{1}{2\pi\sigma_y\omega} \int_{-\infty}^{\infty} \exp\left(-\frac{w^2(y-\mu)^2+\sigma_y^2 y^2}{2\sigma_y^2 w^2}\right)(a_f n + cy)dy \\
&= \frac{1}{2\pi\sigma_y\omega} \int_{-\infty}^{\infty} \exp\left(-\frac{w^2(y^2-2\mu y+\mu^2)+\sigma_y^2 y^2}{2\sigma_y^2 w^2}\right)(a_f n + cy)dy \\
&= \frac{1}{2\pi\sigma_y\omega} \int_{-\infty}^{\infty} \exp\left(-\frac{(w^2+\sigma_y^2)y^2-2w^2\mu y+w^2\mu^2}{2\sigma_y^2 w^2}\right)(a_f n + cy)dy \\
&= \frac{1}{2\pi\sigma_y\omega} \exp\left(-\frac{w^2\mu^2}{2\sigma_y^2 w^2}\right) \int_{-\infty}^{\infty} \exp\left(-\frac{(w^2+\sigma_y^2)y^2-2w^2\mu y}{2\sigma_y^2 w^2}\right)(a_f n + cy)dy \\
&= \frac{1}{2\pi\sigma_y\omega} \exp\left(-\frac{w^2\mu^2}{2\sigma_y^2 w^2}\right) \\
&\times \int_{-\infty}^{\infty} \exp\left(-\frac{y^2-2w^2\mu y/(w^2+\sigma_y^2)+(w^2\mu/(w^2+\sigma_y^2))^2-(w^2\mu/(w^2+\sigma_y^2))^2}{2\sigma_y^2 w^2/(w^2+\sigma_y^2)}\right) \\
&\times (a_f n + cy)dy \\
&= \frac{1}{2\pi\sigma_y\omega} \exp\left(-\frac{w^2\mu^2}{2\sigma_y^2 w^2}\right) \\
&\times \int_{-\infty}^{\infty} \exp\left(-\frac{(y-w^2\mu/(w^2+\sigma_y^2))^2-(w^2\mu/(w^2+\sigma_y^2))^2}{2\sigma_y^2 w^2/(w^2+\sigma_y^2)}\right) \\
&\times (a_f n + cy)dy \\
&= \frac{1}{2\pi\sigma_y\omega} \exp\left(-\frac{w^2\mu^2}{2\sigma_y^2 w^2}\right) \\
&\times \int_{-\infty}^{\infty} \exp\left(-\frac{(y-w^2\mu/(w^2+\sigma_y^2))^2}{2\sigma_y^2 w^2/(w^2+\sigma_y^2)}\right) \exp\left(\frac{w^2\mu^2}{2\sigma_y^2(w^2+\sigma_y^2)}\right) \\
&\times (a_f n + cy)dy \\
&= \frac{1}{2\pi\sigma_y\omega} \exp\left(-\frac{\mu^2}{2\sigma_y^2}\right) \exp\left(\frac{w^2\mu^2}{2\sigma_y^2(w^2+\sigma_y^2)}\right) \\
&\times \int_{-\infty}^{\infty} \exp\left(-\frac{(y-w^2\mu/(w^2+\sigma_y^2))^2}{2\sigma_y^2 w^2/(w^2+\sigma_y^2)}\right) \\
&\times (a_f n + cy)dy \\
&= \frac{1}{2\pi\sigma_y\omega} \exp\left(-\frac{\mu^2}{2(w^2+\sigma_y^2)}\right) \\
&\times \int_{-\infty}^{\infty} \exp\left(-\frac{(y-w^2\mu/(w^2+\sigma_y^2))^2}{2\sigma_y^2 w^2/(w^2+\sigma_y^2)}\right) \\
&\times (a_f n + cy)dy
\end{aligned}
$$

We still need to spend more time doing the integrals:

$$\begin{aligned}
\bar{W}(n,\bar{y}) &= \frac{1}{2\pi\sigma_y\omega}\exp\left(-\frac{\mu^2}{2(w^2+\sigma_y^2)}\right) \\
&\times \frac{\sqrt{2\pi\sigma_y^2 w^2/(w^2+\sigma_y^2)}}{\sqrt{2\pi\sigma_y^2 w^2/(w^2+\sigma_y^2)}}\int_{-\infty}^{\infty}\exp\left(-\frac{(y-w^2\mu/(w^2+\sigma_y^2))^2}{2\sigma_y^2 w^2/(w^2+\sigma_y^2)}\right) \\
&\times (a_f n + cy)dy \\
&= \frac{1}{\sqrt{2\pi(w^2+\sigma_y^2)}}\exp\left(-\frac{\mu^2}{2(w^2+\sigma_y^2)}\right) \\
&\times \frac{1}{\sqrt{2\pi\sigma_y^2 w^2/(w^2+\sigma_y^2)}}\int_{-\infty}^{\infty}\exp\left(-\frac{(y-w^2\mu/(w^2+\sigma_y^2))^2}{2\sigma_y^2 w^2/(w^2+\sigma_y^2)}\right) \\
&\times (a_f n + cy)dy \\
&= \frac{1}{\sqrt{2\pi(w^2+\sigma_y^2)}}\exp\left(-\frac{\mu^2}{2(w^2+\sigma_y^2)}\right)\left(a_f n + \frac{cw^2\mu}{w^2+\sigma_y^2}\right)
\end{aligned}$$

If we substitute $\gamma = 1/(w^2+\sigma_y^2)$ and $\mu = \bar{y}$, then,

$$\bar{W}(n,\bar{y}) = \sqrt{\frac{\gamma}{2\pi}}\exp\left(-\frac{\gamma\bar{y}^2}{2}\right)\left(a_f n + \gamma c w^2 \bar{y}\right)$$

If we write $\bar{y} = \bar{a} + \bar{b}n$, then

$$\begin{aligned}
\beta &= \begin{pmatrix}\partial/\partial\bar{a} \\ \partial/\partial\bar{b}\end{pmatrix}\ln\bar{W}(n,\bar{y}) \\
&= \begin{pmatrix}\partial/\partial\bar{a} \\ \partial/\partial\bar{b}\end{pmatrix}\left(\ln\frac{\gamma}{2\pi} - \frac{\gamma(\bar{a}+\bar{b}n)}{2} + \ln\left(a_f n + \gamma c w^2(\bar{a}+\bar{b}n)\right)\right) \\
&= \begin{pmatrix}\frac{-\gamma}{2} + \frac{\gamma c w^2}{a_f n + \gamma c w^2(\bar{a}+\bar{b}n)} \\ \frac{-\gamma n}{2} + \frac{\gamma c w^2 n}{a_f n + \gamma c w^2(\bar{a}+\bar{b}n)}\end{pmatrix}
\end{aligned}$$

Now, we want to calculate $E(\beta)$ but this is difficult... Instead, we can still solve for $\beta = 0$ for a given value of $n > 0$.

If we just set beta=0, it doesn't work because dlnW/da and dlnW/db are linear functions of each other. Specifically, dlnW/db = n dlnW/da. So we get infinitely many solutions.
What the original Lande papers suggests is that we need to take the average of beta first before we solve for the equilibria. However, this is also problematic because integrating beta across n diverges. I suspect this is because fecundity can be negative.